

The Regular Simplex Is Optimal for Equal-Energy Gaussian Classification

A detailed proof through Gaussian maxima, log-concave doubling, and adaptive tilting

Abstract

We prove that among all collections of $n + 1$ unit vectors in $\mathbb{R}^n$, the vertices of a regular simplex maximize the average probability of correct maximum-likelihood decoding in additive standard Gaussian noise, for every signal strength $\lambda > 0$. The proof first rewrites the decoding probability as an exponential moment of a Gaussian maximum and then adds one common Gaussian coordinate so that the regular simplex becomes a vector of independent standard Gaussians. The central analytic ingredient is a product inequality for one-dimensional log-concave factors with zero Gaussian barycenter. It is proved by a doubling operation, a symmetric Gaussian rectangle inequality, and the central limit theorem. A strictly concave variational problem built from the inverse Mills ratio then constructs adaptive exponential tilts that convert this product inequality into first-order stochastic domination of the relevant Gaussian maximum by the independent maximum.

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1. Statement of the problem and the main theorem

Let $n \geq 1$, put $m = n + 1$, and let

$$x_1, \dots, x_m \in S^{n-1} \subset \mathbb{R}^n.$$

Fix a signal strength $\lambda > 0$. A label I is chosen uniformly from $\{1, \dots, m\}$, an independent Gaussian noise vector $Z \sim N(0, I_n)$ is sampled, and one observes

$$Y = \lambda x_I + Z.$$

The decoder chooses an index maximizing the score $\langle Y, x_i \rangle$. When several indices tie, we choose any measurable tie-breaking rule that always selects one of the maximizing indices. This convention matters only when some of the x_i coincide; the averaged success probability below is independent of the particular tie-breaking rule.

Define

$$\psi_\lambda(x_1, \dots, x_m) = \frac{1}{m} \sum_{j=1}^m \mathbb{P} \left(j = \arg \max_{1 \leq i \leq m} \langle Y, x_i \rangle \mid I = j \right).$$

A regular simplex with m vertices is a collection $\Delta_m = \{v_1, \dots, v_m\} \subset S^{m-2}$ satisfying

$$\langle v_i, v_j \rangle = -\frac{1}{m-1} \quad (i \neq j).$$

Such a simplex is unique up to an orthogonal transformation and a relabeling of its vertices.

Main Theorem. For every $n \geq 1$, every $\lambda > 0$, and every $x_1, \dots, x_{n+1} \in S^{n-1}$,

$$\psi_\lambda(x_1, \dots, x_{n+1}) \leq \psi_\lambda(\Delta_{n+1}).$$

Thus the regular simplex maximizes the average probability of correct maximum-likelihood decoding in every dimension and at every signal-to-noise ratio.

The proof establishes a stronger stochastic statement. After a Gaussian normalization, the regular simplex becomes a vector of independent standard Gaussians. We prove that the maximum arising from any competing code is stochastically no larger than the maximum of those independent Gaussians.

1.1. Roadmap

The argument has four main stages.

1. **Classification becomes a Gaussian maximum.** We rewrite ψ_λ as a positive exponential moment of the maximum of a Gaussian score vector.
2. **The regular simplex becomes independence.** By adjoining one common Gaussian coordinate, every code is transformed into a correlation matrix R , while the regular simplex is transformed into $R = I_m$.
3. **A centered log-concave Gaussian product inequality.** For any Gaussian correlation structure, one-dimensional log-concave factors with zero Gaussian barycenter are positively correlated. The proof uses a doubling operation, a symmetric Gaussian rectangle inequality, and the central limit theorem.
4. **Adaptive exponential tilting.** For a fixed threshold c , a strictly concave finite-dimensional variational problem constructs one tilt per coordinate. These tilts simultaneously center the one-dimensional factors and shift all multivariate thresholds to the same value c . The product inequality then yields the desired Gaussian CDF comparison.

The only convex-analytic theorem used without proof is Prékopa's theorem, stated precisely in Section 2. All other ingredients needed for the main argument are proved below.

2. Conventions and preliminary facts

2.1. Log-concave functions

A function $f : \mathbb{R}^d \rightarrow [0, \infty)$ is **log-concave** if

$$f((1 - \theta)x + \theta y) \geq f(x)^{1-\theta} f(y)^\theta$$

for all $x, y \in \mathbb{R}^d$ and $\theta \in [0, 1]$. Equivalently, one may write $f = e^{-V}$ where $V : \mathbb{R}^d \rightarrow (-\infty, \infty]$ is convex. This extended-valued convention allows f to vanish. In particular, the indicator $\mathbf{1}_K$ of a convex set K is log-concave.

We repeatedly use the following elementary closure properties.

- A product of nonnegative log-concave functions is log-concave.
- If f is log-concave and T is affine, then $f \circ T$ is log-concave.
- Every Gaussian density is log-concave.

The nontrivial closure property we need is preservation under marginalization.

Prékopa's Theorem. Let $F : \mathbb{R}^{p+q} \rightarrow [0, \infty)$ be a measurable log-concave function. Suppose that $\int_{\mathbb{R}^q} F(x, y) dy$ is finite for every x . Then

$$x \mapsto \int_{\mathbb{R}^q} F(x, y) dy$$

is log-concave on $\mathbb{R}^p$.

Two consequences will be important. First, convolving a log-concave function with a Gaussian density produces a log-concave function. Second, if K is convex and W is Gaussian, then the function

$$t \mapsto \mathbb{P}(W + tb \in K)$$

is log-concave for every fixed vector b .

2.2. Standard Gaussian notation

We write

$$\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}, \quad \Phi(t) = \int_{-\infty}^t \phi(u) du,$$

and write γ for standard Gaussian measure on $\mathbb{R}$.

A symmetric positive-semidefinite matrix $R \in \mathbb{R}^{k \times k}$ with diagonal entries equal to 1 is called a **correlation matrix**. If $X \sim N(0, R)$, every coordinate X_i is a standard Gaussian, even when R is singular.

2.3. A monotone covariance identity

We use the following one-dimensional fact. If U, U' are independent and identically distributed real random variables and a, b are both nondecreasing, or both nonincreasing, then

$$2 \operatorname{Cov}(a(U), b(U)) = \mathbb{E}[(a(U) - a(U'))(b(U) - b(U'))] \geq 0.$$

The displayed identity follows by expanding the right-hand side.

3. From maximum-likelihood decoding to a Gaussian maximum

Let

$$G = (\langle x_i, x_j \rangle)_{i,j=1}^m$$

be the Gram matrix of the code. If $Z \sim N(0, I_n)$, define the Gaussian score vector

$$\xi_i = \langle Z, x_i \rangle, \quad \xi = (\xi_1, \dots, \xi_m).$$

Then $\xi \sim N(0, G)$ because

$$\mathbb{E}[\xi_i \xi_j] = \mathbb{E}[\langle Z, x_i \rangle \langle Z, x_j \rangle] = \langle x_i, x_j \rangle.$$

Let

$$M_G = \max_{1 \leq i \leq m} \xi_i.$$

Proposition 3.1 (maximum-likelihood integral identity).

$$\psi_\lambda(x_1, \dots, x_m) = \frac{e^{-\lambda^2/2}}{m} \mathbb{E} e^{\lambda M_G}.$$

Proof. Let ϕ_n be the standard Gaussian density on $\mathbb{R}^n$. Conditional on $I = i$, the observation Y has density

$$p_i(y) = \phi_n(y - \lambda x_i).$$

Because $\|x_i\| = 1$,

$$\begin{aligned}\phi_n(y - \lambda x_i) &= (2\pi)^{-n/2} \exp\left(-\frac{1}{2}\|y - \lambda x_i\|^2\right) \\ &= \phi_n(y) \exp\left(\lambda \langle y, x_i \rangle - \frac{\lambda^2}{2}\right).\end{aligned}$$

Thus the index maximizing $p_i(y)$ is exactly an index maximizing $\langle y, x_i \rangle$.

Let D_i be the measurable decision region assigned to label i . The sets $D_1, \dots, D_m$ form a measurable partition, and for $y \in D_i$, the density $p_i(y)$ equals $\max_j p_j(y)$. Therefore

$$\begin{aligned}m\psi_\lambda &= \sum_{i=1}^m \int_{D_i} p_i(y) dy \\ &= \int_{\mathbb{R}^n} \max_{1 \leq i \leq m} p_i(y) dy \\ &= e^{-\lambda^2/2} \int_{\mathbb{R}^n} \exp\left(\lambda \max_i \langle y, x_i \rangle\right) \phi_n(y) dy \\ &= e^{-\lambda^2/2} \mathbb{E} e^{\lambda M_G}.\end{aligned}$$

Dividing by m proves the proposition. The same calculation shows that the average success probability is unaffected by the chosen tie-breaking rule. $\square$

4. A normalization under which the simplex is independent

Set

$$\alpha = \frac{m-1}{m}, \quad \beta = \frac{m}{m-1} = \alpha^{-1},$$

and define

$$R = \alpha G + \frac{1}{m} J, \quad J = \mathbf{1}\mathbf{1}^\top.$$

Since $G \succeq 0$ and $J \succeq 0$, we have $R \succeq 0$. Also,

$$R_{ii} = \alpha G_{ii} + \frac{1}{m} = \frac{m-1}{m} + \frac{1}{m} = 1.$$

Hence R is a correlation matrix.

The following probabilistic construction explains this transformation. Let

$$B \sim N\left(0, \frac{1}{m-1}\right)$$

be independent of $\xi \sim N(0, G)$, and define

$$X_i = \sqrt{\alpha} (\xi_i + B).$$

Then $X = (X_1, \dots, X_m)$ is centered Gaussian and

$$\text{Cov}(X) = \alpha \left(G + \frac{1}{m-1} J\right) = \alpha G + \frac{1}{m} J = R.$$

Let

$$M_R = \max_i X_i, \quad \mu = \frac{\lambda}{\sqrt{\alpha}} = \lambda \sqrt{\frac{m}{m-1}}.$$

Because the same random variable B is added to all coordinates,

$$M_R = \sqrt{\alpha}(M_G + B), \quad \mu M_R = \lambda(M_G + B).$$

Using independence of B and M_G ,

$$\begin{aligned} e^{-\mu^2/2} \mathbb{E} e^{\mu M_R} &= e^{-\lambda^2/(2\alpha)} \mathbb{E} e^{\lambda M_G} \mathbb{E} e^{\lambda B} \\ &= e^{-\lambda^2 m/[2(m-1)]} e^{\lambda^2/[2(m-1)]} \mathbb{E} e^{\lambda M_G} \\ &= e^{-\lambda^2/2} \mathbb{E} e^{\lambda M_G}. \end{aligned}$$

Proposition 3.1 therefore becomes

$$\psi_\lambda(x_1, \dots, x_m) = \frac{1}{m} e^{-\mu^2/2} \mathbb{E}_{N(0,R)} e^{\mu M_R}.$$

Now consider a regular simplex. Its Gram matrix is

$$G_\Delta = \frac{m}{m-1} \left(I - \frac{1}{m} J \right).$$

Consequently,

$$\alpha G_\Delta + \frac{1}{m} J = I - \frac{1}{m} J + \frac{1}{m} J = I.$$

Thus the regular simplex corresponds exactly to m independent standard Gaussian variables.

It is therefore enough to prove the following stronger statement for every correlation matrix R arising from the normalization above:

$$\mathbb{P}_{N(0,R)} \left(\max_i X_i \leq c \right) \geq \Phi(c)^m \quad (c \in \mathbb{R}).$$

The right-hand side is the distribution function of the maximum of m independent standard Gaussians.

5. Symmetric Gaussian rectangles

The product theorem in the next section is built on a symmetric rectangle inequality. We prove the needed form directly.

Lemma 5.1 (symmetric Gaussian rectangle inequality). Let $V = (V_1, \dots, V_k)$ be a centered Gaussian vector with $\text{Var}(V_i) = 1$ for every i . Then, for all $r_1, \dots, r_k \in [0, \infty]$,

$$\mathbb{P}(|V_i| \leq r_i \text{ for every } i) \geq \prod_{i=1}^k \mathbb{P}(|Z| \leq r_i),$$

where $Z \sim N(0, 1)$.

Proof. We first suppose that the covariance matrix of V is positive definite and proceed by induction on k .

The case $k = 1$ is equality. Assume the result is known in dimension $k - 1$. Let

$$A = \{(v_1, \dots, v_{k-1}) : |v_i| \leq r_i \text{ for } 1 \leq i < k\}.$$

Conditional on $V_k = t$, the vector $(V_1, \dots, V_{k-1})$ has the form

$$Y + bt,$$

where Y is a centered Gaussian vector whose covariance does not depend on t , and $b \in \mathbb{R}^{k-1}$ is fixed. Define

$$q(t) = \mathbb{P}(Y + bt \in A).$$

The set A is symmetric and convex. Since Y and $-Y$ have the same distribution,

$$q(-t) = \mathbb{P}(Y - bt \in A) = \mathbb{P}(-Y - bt \in A) = \mathbb{P}(Y + bt \in A) = q(t).$$

Thus q is even.

The function q is log-concave. Indeed, if ϕ_Y is the Gaussian density of Y , then

$$q(t) = \int_{\mathbb{R}^{k-1}} \mathbf{1}_A(y + bt) \phi_Y(y) dy.$$

The integrand is jointly log-concave in (t, y) : the indicator is the indicator of a convex set after an affine change of variables, and ϕ_Y is log-concave. Prékopa's theorem therefore implies that q is log-concave.

Every even log-concave function on $\mathbb{R}$ is nonincreasing on $[0, \infty)$. Hence both

$$s \mapsto q(s) \quad \text{and} \quad s \mapsto \mathbf{1}_{\{s \leq r_k\}}$$

are nonincreasing functions of $s = |V_k|$. By the monotone covariance identity from Section 2.3,

$$\mathbb{E} \left[q(V_k) \mathbf{1}_{\{|V_k| \leq r_k\}} \right] \geq \mathbb{E} q(V_k) \mathbb{P}(|V_k| \leq r_k).$$

The left side is the probability of the full k -dimensional rectangle. The first factor on the right is

$$\mathbb{E} q(V_k) = \mathbb{P}(|V_i| \leq r_i \text{ for } 1 \leq i < k).$$

Since V_k is standard Gaussian, the second factor is $\mathbb{P}(|Z| \leq r_k)$. Applying the induction hypothesis to the first $k-1$ coordinates proves the result.

If the covariance matrix is singular, replace it by

$$R_\varepsilon = (1 - \varepsilon)R + \varepsilon I, \quad \varepsilon > 0.$$

The inequality holds for R_ε . The corresponding Gaussian vectors converge in distribution to V as $\varepsilon \downarrow 0$, and the boundary of a finite symmetric rectangle has Gaussian probability zero because each coordinate has a continuous standard normal distribution. Passing to the limit completes the proof.

□

The rectangle inequality has a useful functional form.

Corollary 5.2 (even log-concave factors). Let V be as in Lemma 5.1. If $g_1, \dots, g_k : \mathbb{R} \rightarrow [0, \infty)$ are bounded, even, and log-concave, then

$$\mathbb{E} \prod_{i=1}^k g_i(V_i) \geq \prod_{i=1}^k \int_{\mathbb{R}} g_i d\gamma.$$

Proof. For each i and each level $s \geq 0$, the superlevel set

$$\{x : g_i(x) > s\}$$

is a convex symmetric subset of $\mathbb{R}$, hence is a centered interval, possibly empty or unbounded. The layer-cake formula and Tonelli's theorem give

$$\begin{aligned} \mathbb{E} \prod_i g_i(V_i) &= \int_{[0, \infty)^k} \mathbb{P}(g_i(V_i) > s_i \text{ for all } i) ds_1 \cdots ds_k \\ &\geq \int_{[0, \infty)^k} \prod_i \mathbb{P}(g_i(Z) > s_i) ds_1 \cdots ds_k \\ &= \prod_i \mathbb{E} g_i(Z). \end{aligned}$$

The inequality in the second line is Lemma 5.1. $\square$

6. A centered log-concave Gaussian product theorem

We now prove the analytic result that will later be applied to tilted half-lines.

Theorem 6.1 (centered log-concave Gaussian product inequality). Let R be any $k \times k$ correlation matrix, let $X \sim N(0, R)$, and let $f_1, \dots, f_k$ be bounded, nonnegative log-concave functions with positive Gaussian integrals. Assume that each factor has zero Gaussian barycenter:

$$\int_{\mathbb{R}} x f_i(x) d\gamma(x) = 0.$$

Then

$$\mathbb{E} \prod_{i=1}^k f_i(X_i) \geq \prod_{i=1}^k \int_{\mathbb{R}} f_i d\gamma.$$

We divide the proof into several steps.

6.1. Normalization and the doubling operator

Normalize each factor by setting

$$h_i = \frac{f_i}{\int f_i d\gamma}.$$

Then

$$\int h_i d\gamma = 1, \quad \int x h_i(x) d\gamma(x) = 0.$$

The probability measure $h_i d\gamma$ therefore has mean zero.

For a bounded nonnegative function h , define its **doubling transform** by

$$(\mathcal{D}h)(u) = \int_{\mathbb{R}} h\left(\frac{u+v}{\sqrt{2}}\right) h\left(\frac{u-v}{\sqrt{2}}\right) d\gamma(v).$$

We record four properties.

(a) Log-concavity. If h is log-concave, then $\mathcal{D}h$ is log-concave. With respect to Lebesgue measure, the integrand is

$$(u, v) \mapsto h\left(\frac{u+v}{\sqrt{2}}\right) h\left(\frac{u-v}{\sqrt{2}}\right) \phi(v).$$

It is jointly log-concave, so Prékopa's theorem applies after integrating over v .

(b) Preservation of mass. If $\int h d\gamma = 1$, then

$$\int \mathcal{D}h d\gamma = 1.$$

Indeed, the orthogonal change of variables

$$y = \frac{u+v}{\sqrt{2}}, \quad y' = \frac{u-v}{\sqrt{2}}$$

preserves the product Gaussian measure $d\gamma(u)d\gamma(v)$, so

$$\begin{aligned} \int \mathcal{D}h(u) d\gamma(u) &= \iint h\left(\frac{u+v}{\sqrt{2}}\right) h\left(\frac{u-v}{\sqrt{2}}\right) d\gamma(v)d\gamma(u) \\ &= \iint h(y)h(y') d\gamma(y)d\gamma(y') = 1. \end{aligned}$$

(c) Preservation of the barycenter. If $\int xh(x) d\gamma(x) = 0$, then

$$\int u \mathcal{D}h(u) d\gamma(u) = 0.$$

Under the same orthogonal change of variables, $u = (y + y')/\sqrt{2}$, and the integral becomes

$$\frac{1}{\sqrt{2}} \iint (y + y')h(y)h(y') d\gamma(y)d\gamma(y') = 0.$$

(d) Probabilistic meaning. If Y, Y' are independent with common law $h d\gamma$, then $\mathcal{D}h d\gamma$ is the law of

$$\frac{Y + Y'}{\sqrt{2}}.$$

This follows from the same orthogonal change of variables and is the reason for the name “doubling transform.”

6.2. The key doubling inequality

For bounded functions $q_1, \dots, q_k$, define

$$\mathcal{Z}_R(q_1, \dots, q_k) = \mathbb{E}_{N(0, R)} \prod_{i=1}^k q_i(X_i).$$

Let X, X' be independent copies of $N(0, R)$, and put

$$U = \frac{X + X'}{\sqrt{2}}, \quad V = \frac{X - X'}{\sqrt{2}}.$$

Because (X, X') is jointly Gaussian and the transformation is orthogonal in the copy index, U and V are independent, and each has distribution $N(0, R)$.

Therefore

$$\mathcal{Z}_R(h_1, \dots, h_k)^2 = \mathbb{E}_{U,V} \prod_{i=1}^k h_i \left(\frac{U_i + V_i}{\sqrt{2}} \right) h_i \left(\frac{U_i - V_i}{\sqrt{2}} \right).$$

Fix $U = u$ and define, as a function of one real variable v ,

$$g_{i,u_i}(v) = h_i \left(\frac{u_i + v}{\sqrt{2}} \right) h_i \left(\frac{u_i - v}{\sqrt{2}} \right).$$

This function is even, bounded, and log-concave. Corollary 5.2, applied to the conditional expectation over V , gives

$$\begin{aligned} \mathbb{E}_V \prod_i g_{i,u_i}(V_i) &\geq \prod_i \int g_{i,u_i}(v) d\gamma(v) \\ &= \prod_i (\mathcal{D}h_i)(u_i). \end{aligned}$$

Averaging over U yields the key inequality

$$\mathcal{Z}_R(h_1, \dots, h_k)^2 \geq \mathcal{Z}_R(\mathcal{D}h_1, \dots, \mathcal{D}h_k).$$

6.3. Iteration and the central limit theorem

Define recursively

$$h_{i,0} = h_i, \quad h_{i,r+1} = \mathcal{D}h_{i,r},$$

and write

$$Z_r = \mathcal{Z}_R(h_{1,r}, \dots, h_{k,r}).$$

Iterating the doubling inequality gives

$$Z_0^{2^r} \geq Z_r \quad (r \geq 0).$$

Let $\nu_i = h_i d\gamma$, and let $Y_{i,1}, Y_{i,2}, \dots$ be independent random variables with law ν_i . Property (d) of the doubling transform implies that $h_{i,r} d\gamma$ is the law of

$$S_{i,r} = 2^{-r/2} \sum_{\ell=1}^{2^r} Y_{i,\ell}.$$

The law ν_i has mean zero. Because h_i is bounded, all its moments are finite. Its variance σ_i^2 is strictly positive: ν_i is absolutely continuous with respect to Gaussian measure and therefore cannot be a point mass. The central limit theorem gives

$$S_{i,r} \implies N(0, \sigma_i^2).$$

Assume first that R is positive definite. Its density relative to the product standard Gaussian measure $\gamma^{\otimes k}$ is

$$L_R(x) = (\det R)^{-1/2} \exp\left(-\frac{1}{2} x^\top (R^{-1} - I)x\right).$$

Fix any $L > 0$. The function L_R is continuous and strictly positive, so

$$c_{R,L} = \min_{x \in [-L,L]^k} L_R(x) > 0.$$

Consequently,

$$\begin{aligned} Z_r &= \int_{\mathbb{R}^k} L_R(x) \prod_i h_{i,r}(x_i) d\gamma^{\otimes k}(x) \\ &\geq c_{R,L} \prod_i \int_{-L}^L h_{i,r}(x) d\gamma(x) \\ &= c_{R,L} \prod_i \mathbb{P}(|S_{i,r}| \leq L). \end{aligned}$$

By the central limit theorem, each factor converges to

$$\mathbb{P}(|\sigma_i Z| \leq L) > 0.$$

Thus

$$\liminf_{r \rightarrow \infty} Z_r > 0.$$

If $Z_0 < 1$, then $Z_0^{2^r} \rightarrow 0$, and the inequality $Z_r \leq Z_0^{2^r}$ would force $Z_r \rightarrow 0$, contradicting the positive lower bound. Therefore $Z_0 \geq 1$. Undoing the normalization of the f_i proves the theorem when R is positive definite.

6.4. Singular correlation matrices

For a singular correlation matrix R , set

$$R_\varepsilon = (1 - \varepsilon)R + \varepsilon I, \quad \varepsilon > 0.$$

This is a positive-definite correlation matrix, so the theorem holds for R_ε .

To pass to the limit, let $X \sim N(0, R)$ and $Z \sim N(0, I)$ be independent, and define

$$X^{(\varepsilon)} = \sqrt{1 - \varepsilon} X + \sqrt{\varepsilon} Z.$$

Then $X^{(\varepsilon)} \sim N(0, R_\varepsilon)$ and $X^{(\varepsilon)} \rightarrow X$ almost surely.

A bounded one-dimensional log-concave function is continuous in the interior of its support and can be discontinuous only at the two endpoints of that support. Each coordinate X_i has a continuous standard Gaussian distribution, so almost surely X_i is a continuity point of f_i . Hence

$$\prod_i f_i(X_i^{(\varepsilon)}) \rightarrow \prod_i f_i(X_i) \quad \text{almost surely.}$$

The product is uniformly bounded, so dominated convergence applies. Letting $\varepsilon \downarrow 0$ completes the proof of Theorem 6.1. $\square$

6.5. Why the doubling proof works

The centering condition is used only in the central limit step: repeated doubling forms normalized sums without producing a drifting mean. The symmetric rectangle inequality handles the dependence in R after conditioning on the averaged copy U . Iteration drives each one-dimensional factor toward a Gaussian fixed point, while the multivariate partition function cannot collapse to zero. This forces its original value to be at least the independent product.

7. The inverse Mills map and a concave primitive

The adaptive tilt requires a particular change of variables built from the inverse Mills ratio. Define

$$r(s) = \frac{\phi(s)}{\Phi(s)}, \quad H(s) = s + r(s).$$

7.1. Monotonicity and range of H

Differentiating r gives

$$\begin{aligned} r'(s) &= \frac{\phi'(s)\Phi(s) - \phi(s)\Phi'(s)}{\Phi(s)^2} \\ &= \frac{-s\phi(s)\Phi(s) - \phi(s)^2}{\Phi(s)^2} \\ &= -r(s)(s + r(s)) \\ &= -r(s)H(s). \end{aligned}$$

Also,

$$H(s)\Phi(s) = s\Phi(s) + \phi(s).$$

Since

$$\frac{d}{ds}(s\Phi(s) + \phi(s)) = \Phi(s)$$

and the expression tends to 0 as $s \rightarrow -\infty$, we have

$$H(s)\Phi(s) = \int_{-\infty}^s \Phi(u) du > 0.$$

Thus $H(s) > 0$.

For later use, we spell out the truncated-normal calculation. Integration by parts gives

$$\int_{-\infty}^s z\phi(z) dz = -\phi(s)$$

and

$$\int_{-\infty}^s z^2\phi(z) dz = \Phi(s) - s\phi(s).$$

Therefore a standard Gaussian conditioned on $\{Z \leq s\}$ has mean $-r(s)$ and variance

$$1 - sr(s) - r(s)^2.$$

Consequently,

$$H'(s) = 1 + r'(s) = 1 - r(s)H(s) = 1 - sr(s) - r(s)^2 > 0.$$

The strict inequality holds because a Gaussian conditioned on a nontrivial half-line is not almost surely constant. Thus H is strictly increasing.

As $s \rightarrow +\infty$, $r(s) \rightarrow 0$, so $H(s) \rightarrow +\infty$. As $s \rightarrow -\infty$, both $\int_{-\infty}^s \Phi(u) du$ and $\Phi(s)$ tend to zero, and l'Hôpital's rule gives

$$H(s) = \frac{\int_{-\infty}^s \Phi(u) du}{\Phi(s)} \rightarrow \frac{\Phi(s)}{\phi(s)} = \frac{1}{r(s)} \rightarrow 0.$$

The last limit also follows from the Mills estimates proved below. Therefore

$$H : \mathbb{R} \rightarrow (0, \infty)$$

is a strictly increasing bijection.

7.2. Definition and concavity of the primitive

Define $\mathcal{F} : (0, \infty) \rightarrow \mathbb{R}$ by

$$\mathcal{F}(H(s)) = \log \Phi(s) + \frac{1}{2}r(s)^2.$$

This is well-defined because H is bijective.

Differentiating with respect to s ,

$$\begin{aligned} \frac{d}{ds} \left(\log \Phi(s) + \frac{1}{2}r(s)^2 \right) &= r(s) + r(s)r'(s) \\ &= r(s)(1 - r(s)H(s)) \\ &= r(s)H'(s). \end{aligned}$$

Hence

$$\mathcal{F}'(H(s)) = r(s) > 0.$$

Differentiating once more,

$$\mathcal{F}''(H(s)) = \frac{r'(s)}{H'(s)} < 0.$$

Thus $\mathcal{F}$ is strictly increasing and strictly concave.

As $s \rightarrow +\infty$, $\Phi(s) \rightarrow 1$ and $r(s) \rightarrow 0$, so

$$\mathcal{F}(H(s)) \rightarrow 0.$$

Because $\mathcal{F}$ is increasing, it follows that

$$\mathcal{F}(y) < 0 \quad (y > 0).$$

We also need the behavior at the left endpoint. Put $s = -t$ with $t > 0$, and write $Q(t) = \Phi(-t)$. The elementary Mills inequalities

$$\frac{t}{1+t^2}\phi(t) \leq Q(t) \leq \frac{\phi(t)}{t}$$

imply

$$r(-t) = \frac{\phi(t)}{Q(t)} \leq t + \frac{1}{t}.$$

Therefore

$$\begin{aligned} \mathcal{F}(H(-t)) &= \log Q(t) + \frac{1}{2}r(-t)^2 \\ &\leq -\frac{t^2}{2} - \log t - \frac{1}{2}\log(2\pi) + \frac{1}{2}\left(t + \frac{1}{t}\right)^2 \\ &= 1 + \frac{1}{2t^2} - \log t - \frac{1}{2}\log(2\pi), \end{aligned}$$

which tends to $-\infty$. Hence

$$\mathcal{F}(y) \rightarrow -\infty \quad \text{as } y \downarrow 0.$$

For completeness, the upper Mills bound follows from

$$Q(t) = \int_t^\infty \phi(u) du \leq \frac{1}{t} \int_t^\infty u\phi(u) du = \frac{\phi(t)}{t}.$$

Integration by parts gives

$$Q(t) = \frac{\phi(t)}{t} - \int_t^\infty \frac{\phi(u)}{u^2} du \geq \frac{\phi(t)}{t} - \frac{Q(t)}{t^2},$$

which rearranges to the lower Mills bound.

8. The adaptive-tilt variational problem

Fix a threshold $c \in \mathbb{R}$. Recall that

$$R = \alpha G + \frac{1}{m} J, \quad \alpha = \frac{m-1}{m}.$$

For $q \in \mathbb{R}$ and $v \in \mathbb{R}^n$, define

$$\mathcal{J}_c(q, v) = \sum_{i=1}^m \mathcal{F}(q + \alpha \langle x_i, v \rangle) - \frac{\alpha}{2} \|v\|^2 - \frac{m}{2} (q - c)^2$$

on the open convex domain

$$\mathcal{D} = \{(q, v) : q + \alpha \langle x_i, v \rangle > 0 \text{ for all } i\}.$$

The point $(1, 0)$ belongs to $\mathcal{D}$, so the domain is nonempty.

8.1. Existence and uniqueness of the maximizer

The function $\mathcal{J}_c$ is strictly concave. Each term involving $\mathcal{F}$ is concave because $\mathcal{F}$ is concave and its argument is affine. The negative quadratic term

$$-\frac{\alpha}{2} \|v\|^2 - \frac{m}{2} (q - c)^2$$

is strictly concave.

The function also tends to $-\infty$ at every way of escaping the domain. If a bounded sequence in $\mathcal{D}$ approaches the boundary, at least one argument of $\mathcal{F}$ tends to 0, and that summand tends to $-\infty$. If $\|(q, v)\| \rightarrow \infty$, then $\mathcal{F} \leq 0$ and the negative quadratic terms tend to $-\infty$.

For a completely formal compactness argument, choose one point $(q_0, v_0) \in \mathcal{D}$. The superlevel set

$$\{(q, v) \in \mathcal{D} : \mathcal{J}_c(q, v) \geq \mathcal{J}_c(q_0, v_0)\}$$

is bounded and stays a positive distance from the boundary of $\mathcal{D}$. Its closure is therefore a compact subset of $\mathcal{D}$, on which the continuous function $\mathcal{J}_c$ attains its maximum. Strict concavity makes this maximizer unique. Denote it by

$$(q_*, v_*) \in \mathcal{D}.$$

8.2. The stationarity equations

Set

$$y_i = q_* + \alpha \langle x_i, v_* \rangle > 0.$$

Because H maps $\mathbb{R}$ bijectively onto $(0, \infty)$, there is a unique $s_i \in \mathbb{R}$ such that

$$y_i = H(s_i).$$

Define

$$a_i = r(s_i) = \mathcal{F}'(y_i) > 0.$$

Differentiating $\mathcal{J}_c$ with respect to v and using stationarity gives

$$0 = \alpha \sum_{i=1}^m a_i x_i - \alpha v_*,$$

so

$$v_* = \sum_{i=1}^m a_i x_i.$$

Differentiating with respect to q gives

$$0 = \sum_{i=1}^m a_i - m(q_* - c),$$

so

$$q_* = c + \bar{a}, \quad \bar{a} = \frac{1}{m} \sum_{i=1}^m a_i.$$

Let $a = (a_1, \dots, a_m)^\top$. Since $R = \alpha G + J/m$,

$$\begin{aligned} (Ra)_i &= \alpha \sum_{j=1}^m \langle x_i, x_j \rangle a_j + \bar{a} \\ &= \alpha \left\langle x_i, \sum_{j=1}^m a_j x_j \right\rangle + \bar{a} \\ &= \alpha \langle x_i, v_* \rangle + \bar{a}. \end{aligned}$$

Therefore

$$y_i = q_* + \alpha \langle x_i, v_* \rangle = c + (Ra)_i.$$

On the other hand, $y_i = H(s_i) = s_i + a_i$. Hence the vector identity

$$s + a - Ra = c\mathbf{1}$$

holds.

This is the key output of the variational problem: after the Cameron–Martin shift by Ra , all coordinate thresholds become equal to c .

8.3. Lower bound for the variational value

By definition of $\mathcal{F}$,

$$\mathcal{F}(y_i) = \log \Phi(s_i) + \frac{1}{2} a_i^2.$$

Furthermore,

$$\begin{aligned} a^\top R a &= \alpha a^\top G a + \frac{1}{m} \left(\sum_i a_i \right)^2 \\ &= \alpha \left\| \sum_i a_i x_i \right\|^2 + m \bar{a}^2 \\ &= \alpha \|v_*\|^2 + m(q_* - c)^2. \end{aligned}$$

It follows that

$$\mathcal{J}_c(q_*, v_*) = \sum_{i=1}^m \log \Phi(s_i) + \frac{1}{2} a^\top (I - R) a.$$

Now evaluate $\mathcal{J}_c$ at the admissible point

$$q = H(c) = c + r(c), \quad v = 0.$$

Using the definition of $\mathcal{F}$,

$$\begin{aligned} \mathcal{J}_c(H(c), 0) &= m \mathcal{F}(H(c)) - \frac{m}{2} r(c)^2 \\ &= m \left(\log \Phi(c) + \frac{1}{2} r(c)^2 \right) - \frac{m}{2} r(c)^2 \\ &= m \log \Phi(c). \end{aligned}$$

Since (q_*, v_*) is the maximizer,

$$\sum_{i=1}^m \log \Phi(s_i) + \frac{1}{2} a^\top (I - R) a \geq m \log \Phi(c).$$

9. Exponentially tilted half-lines and the Gaussian CDF bound

Let $X \sim N(0, R)$. For each i , define

$$f_i(z) = e^{a_i z} \mathbf{1}_{\{z \leq s_i + a_i\}}.$$

Because $a_i > 0$, the function is bounded: on its support,

$$e^{a_i z} \leq e^{a_i(s_i + a_i)}.$$

It is nonzero and log-concave because its logarithm is affine on a half-line and equals $-\infty$ outside that half-line.

Completing the square gives its Gaussian mass:

$$\begin{aligned} \int f_i(z) d\gamma(z) &= \int_{-\infty}^{s_i + a_i} e^{a_i z} \phi(z) dz \\ &= e^{a_i^2/2} \int_{-\infty}^{s_i} \phi(u) du \\ &= e^{a_i^2/2} \Phi(s_i). \end{aligned}$$

Its Gaussian barycenter is

$$\begin{aligned} \int z f_i(z) d\gamma(z) &= e^{a_i^2/2} \int_{-\infty}^{s_i} (u + a_i) \phi(u) du \\ &= e^{a_i^2/2} (a_i \Phi(s_i) - \phi(s_i)) \\ &= 0, \end{aligned}$$

because $a_i = r(s_i) = \phi(s_i)/\Phi(s_i)$.

Theorem 6.1 therefore yields

$$\begin{aligned} \mathbb{E} \left[e^{a^\top X} \mathbf{1}_{\{X_i \leq s_i + a_i \ \forall i\}} \right] &\geq \prod_{i=1}^m e^{a_i^2/2} \Phi(s_i) \\ &= e^{\|a\|^2/2} \prod_{i=1}^m \Phi(s_i). \end{aligned}$$

We now use a Gaussian shift identity.

Lemma 9.1 (Cameron–Martin shift in finite dimensions). Let $X \sim N(0, R)$, where R may be singular. For every $a \in \mathbb{R}^m$ and every Borel set $B \subseteq \mathbb{R}^m$,

$$\mathbb{E} \left[e^{a^\top X} \mathbf{1}_{\{X \in B\}} \right] = e^{a^\top R a / 2} \mathbb{P}(X + Ra \in B).$$

Proof. Choose a matrix T and a standard Gaussian vector Z such that $X = TZ$ and $TT^\top = R$. Then

$$a^\top X = (T^\top a)^\top Z.$$

The standard Gaussian density satisfies

$$e^{b^\top z} \phi_d(z) = e^{\|b\|^2/2} \phi_d(z - b).$$

Applying this with $b = T^\top a$ and changing variables gives

$$\begin{aligned} \mathbb{E} \left[e^{a^\top X} \mathbf{1}_{\{X \in B\}} \right] &= e^{\|T^\top a\|^2/2} \mathbb{P}(T(Z + T^\top a) \in B) \\ &= e^{a^\top R a / 2} \mathbb{P}(X + Ra \in B). \end{aligned}$$

This proof does not require R to be invertible. $\square$

Apply the lemma to

$$B = \{x \in \mathbb{R}^m : x_i \leq s_i + a_i \text{ for all } i\}.$$

We obtain

$$\begin{aligned} \mathbb{E} \left[e^{a^\top X} \mathbf{1}_{\{X_i \leq s_i + a_i \ \forall i\}} \right] \\ = e^{a^\top R a / 2} \mathbb{P}(X \leq s + a - Ra). \end{aligned}$$

The variational identity $s + a - Ra = c\mathbf{1}$ therefore gives

$$e^{a^\top R a / 2} \mathbb{P}(X_i \leq c \text{ for all } i) \geq e^{\|a\|^2/2} \prod_i \Phi(s_i).$$

The right-hand side is strictly positive, so the probability on the left is positive and taking logarithms is legitimate. We obtain

$$\log \mathbb{P}(X_i \leq c \text{ for all } i) \geq \sum_i \log \Phi(s_i) + \frac{1}{2} a^\top (I - R) a.$$

The lower bound from Section 8.3 now yields

$$\log \mathbb{P}(X_i \leq c \text{ for all } i) \geq m \log \Phi(c).$$

Exponentiating,

$$\mathbb{P}_{N(0,R)} \left(\max_i X_i \leq c \right) \geq \Phi(c)^m \quad (c \in \mathbb{R}).$$

If $G_1, \dots, G_m$ are independent standard Gaussians, then

$$\mathbb{P} \left(\max_i G_i \leq c \right) = \Phi(c)^m.$$

Hence

$$\max_i X_i \leq_{\text{st}} \max_i G_i,$$

where $\leq_{\text{st}}$ denotes first-order stochastic domination.

10. Completion of the main theorem

For every $\mu > 0$, the function $x \mapsto e^{\mu x}$ is increasing. The Gaussian maxima under consideration have finite exponential moments of every order. Therefore stochastic domination, first applied to bounded truncations of $e^{\mu x}$ and then passed to the limit by monotone convergence, implies

$$\mathbb{E}_{N(0,R)} e^{\mu \max_i X_i} \leq \mathbb{E} e^{\mu \max_i G_i},$$

where the G_i are independent standard Gaussians.

Returning to the normalization in Section 4,

$$\psi_\lambda(x_1, \dots, x_m) = \frac{1}{m} e^{-\mu^2/2} \mathbb{E}_{N(0,R)} e^{\mu \max_i X_i}$$

and therefore

$$\psi_\lambda(x_1, \dots, x_m) \leq \frac{1}{m} e^{-\mu^2/2} \mathbb{E} e^{\mu \max_i G_i}.$$

The expression on the right is exactly the value associated with the regular simplex, because the regular simplex is transformed into the independent correlation matrix I_m . Thus

$$\psi_\lambda(x_1, \dots, x_m) \leq \psi_\lambda(\Delta_m).$$

This proves the Main Theorem. $\square$

11. A useful explicit formula for the simplex value

Although it is not needed for the comparison proof, the optimum can be written as a one-dimensional Gaussian integral. Let $G_1, \dots, G_m$ be independent standard Gaussians, let

$$\bar{G} = \frac{1}{m} \sum_{i=1}^m G_i,$$

and put

$$\mu = \lambda \sqrt{\frac{m}{m-1}}.$$

A Gaussian score vector with the regular-simplex covariance has the representation

$$\left(\sqrt{\frac{m}{m-1}} (G_i - \bar{G}) \right)_{i=1}^m.$$

Indeed, its diagonal entries are 1 and its off-diagonal entries are $-1/(m-1)$.

Condition on label 1 being transmitted. Correct decoding means that, for every $j \neq 1$,

$$\lambda + \sqrt{\frac{m}{m-1}} (G_1 - \bar{G}) \geq -\frac{\lambda}{m-1} + \sqrt{\frac{m}{m-1}} (G_j - \bar{G}).$$

After canceling $\bar{G}$ and rearranging, this is equivalent to

$$G_j \leq G_1 + \mu \quad (2 \leq j \leq m).$$

Conditioning on G_1 therefore gives

$$\boxed{\psi_\lambda(\Delta_m) = \mathbb{E}[\Phi(G_1 + \mu)^{m-1}].}$$

In the original notation $m = n + 1$, this becomes

$$\boxed{\psi_\lambda(\Delta_{n+1}) = \mathbb{E} \left[\Phi \left(G + \lambda \sqrt{\frac{n+1}{n}} \right)^n \right], \quad G \sim N(0, 1).}$$